

Weighted-Schur growing affine-root-union origin gap

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope: a cardinality-uniform affine origin gap

R-155 permits source-target reuse in the canonical fixed-law affine chart. Let $\mathcal{P}$ be any finite set of distinct nonzero retained side-16 dual-lattice momentum classes modulo sign. Retain every doubled class $[2p]$ and allow $\mathcal{P} \cap 2\mathcal{P} \neq \emptyset$. Define the centered zero-control Gaussian law once on $\mathcal{P} \cup 2\mathcal{P}$, with one raw innovation $\xi_{[r]}$ for each antipodal class, and set

$$h_{[2p]} = t_p H_p \xi_{[p]}, \quad H_p \in \mathbb{R}^{12 \times 12}. \quad (1.1)$$

The feedback functions in (1.1) are fixed during differentiation. A reused target is not replaced by an already shifted state.

Fix the hash-pinned A1 symbol, the A7 positive floor, a finite cutoff, exact nonaliased torus integration, and a common real-even covariance-matched scalar multiplier m_Λ with $|m_\Lambda(k)| \leq 1$, identical on all real components. At the centered stationary A1/A6/A7 origin, R-160 proves first the continuous radial certificate

$$\boxed{D^2 \mathcal{A}(0)[H, H] > \frac{19}{160} \sum_{[p] \in \mathcal{P}} \|H_p\|_{\text{HS}}^2, \quad H \neq 0.} \quad (1.2)$$

It then uses the exact side-16 lattice shells to sharpen (1.2) to

$$\boxed{D^2 \mathcal{A}(0)[H, H] > \frac{4}{25} \sum_{[p] \in \mathcal{P}} \|H_p\|_{\text{HS}}^2, \quad H \neq 0.} \quad (1.3)$$

Both constants are independent of the finite cardinality of $\mathcal{P}$, the number of factor-three components, and their finite path lengths. The stronger margin in (1.3) is

$$\frac{4}{25} - \frac{1}{10} = \frac{3}{50}. \quad (1.4)$$

This is an origin-Hessian theorem for the R-155 fixed-law chart. It gives no cardinality-uniform nonzero radius, nonzero or realised-past estimate, shifted-state recursive or revisited feedback theorem, arbitrary predictable nonlinearity, finite-amplitude convexity, cutoff or floor removal, complete historical low owner, T-050 or A13 closure, Nelson theorem, interacting measure, phase or PDE verdict, or Sector-A closure.

2. Correct authority symbol and the adverse endpoint majorant

The exact registered radial lower symbol is

$$f(x) = x^2 - \frac{4626377063}{5000000000}x + \frac{5020336473}{10000000000}, \quad x = |p|^2. \quad (2.1)$$

Its discriminant is

$$-\frac{28800000000947494031}{2500000000000000000} < 0. \quad (2.2)$$

An exploratory calculation initially used

$$f_0(x) = x^2 - \frac{953}{1000}x + \frac{5013}{10000}. \quad (2.3)$$

That expression is not the authority symbol. The hostile audit caught the error before registration. It is only a strict minorant because

$$f(x) - f_0(x) = \frac{138622937}{5000000000}x + \frac{7336473}{10000000000} > 0 \quad (x \geq 0). \quad (2.4)$$

Every certificate below is recomputed from (2.1); no sign list or fraction from the discarded scratch calculation is reused.

The inherited exact production data are

$$V = 4096, \quad H_6 = \frac{14166000000000}{4000000000001} < \frac{7083}{2000}, \quad (2.5)$$

and hence

$$A = \frac{624(7083/2000)}{4096} = \frac{276237}{512000}, \quad D = \frac{2256(7083/2000)}{4096} = \frac{998703}{512000}. \quad (2.6)$$

Put

$$F_{14}(x) = f(x)f(4x), \quad F_{936}(x) = f(9x)f(36x), \quad (2.7)$$

$$L(x) = \frac{Ax}{F_{14}(x)}, \quad E(x) = \frac{Dx}{\sqrt{F_{14}(x)F_{936}(x)}}. \quad (2.8)$$

R-155 derives the complete mixed covariance-normal endpoint Hessian from one scalar action. Multiplication by two is injective on the torsion-free dual lattice, distinct raw sources remain independent, and factor-two source-target reuse creates no endpoint edge. The only distinct response ratios are 1/3 and 3. Thus each endpoint/loss component is a finite factor-three path. Orient one component as

$$p_j = 3^j p_0, \quad x_j = |p_j|^2 = 9^j x_0. \quad (2.9)$$

For $u_j = \|H_{p_j}\|_{\text{HS}}$, the exact block bounds imply the adverse scalar comparison

$$D^2\mathcal{R}(0)[H, H] > -\sum_j L(x_j)u_j^2 - 2\sum_j E(x_j)u_j u_{j+1}. \quad (2.10)$$

Equation (2.10) is a lower comparison. The functions L, E are not the exact signed tensor spectrum and do not bound the positive endpoint operator norm. The legal reverse is the adjoint of the same off-diagonal block, not a second edge payment.

The Cameron-Martin source Hessian is

$$D^2\mathcal{A}_{\text{CM}}(0)[H, H] = \frac{9}{10} \sum_j u_j^2. \quad (2.11)$$

The full terminal sixth-power Hessian can be dense, but it is one global positive-semidefinite form. It is dropped wholesale. No sparsity or diagonal allocation is asserted for its mixed entries.

3. Weighted Schur payment on every finite path

Choose the geometric Schur weights $d_j = 4^{-j}$. The exact identity

$$\frac{1}{4}a^2 + 4b^2 - 2ab = \left(\frac{a}{2} - 2b\right)^2 \geq 0 \quad (3.1)$$

gives, for every path edge,

$$2E(x_j)u_ju_{j+1} \leq E(x_j) \left(\frac{1}{4}u_j^2 + 4u_{j+1}^2\right). \quad (3.2)$$

Consequently the adverse row losses are

$$\begin{aligned} \text{root:} \quad & L(x_0) + \frac{1}{4}E(x_0), \\ \text{interior:} \quad & L(x_j) + 4E(x_j/9) + \frac{1}{4}E(x_j), \\ \text{terminal:} \quad & L(x_m) + 4E(x_m/9), \\ \text{singleton:} \quad & L(x_0). \end{aligned} \quad (3.3)$$

This is the block Young form of a weighted Schur or Collatz estimate. It is an inequality in the original summed Hilbert-Schmidt norm; no condition number for a long diagonal similarity is inserted.

The graph has degree at most two because a class has at most one factor-three parent and one child. Its norm strictly grows along the orientation, so an exact-torus component cannot cycle. Fixed cutoff makes every admitted path finite. Summing (3.2) proves the same row constant for every finite path length and for every finite number of components.

4. Continuous half-line certificate and the 19/160 gap

The elementary bound $\pi > 31/10$ gives every nonzero side-16 dual mode

$$x = \left(\frac{2\pi}{16}\right)^2 |n|^2 > \frac{3}{20}. \quad (4.1)$$

Let $t_c = 25/32$ and define

$$G_c(x) = t_c F_{14}(x) - Ax, \quad (4.2)$$

$$P_c(x) = G_c(x)^2 F_{936}(x) - \frac{D^2 x^2 F_{14}(x)}{16}. \quad (4.3)$$

The sign guard $G_c > 0$ makes squaring reversible. Therefore

$$G_c(x) > 0, \quad P_c(x) > 0 \quad \implies \quad L(x) + \frac{1}{4}E(x) < \frac{25}{32}. \quad (4.4)$$

The primary exact Sturm implementation finds no root of G_c or P_c on $[3/20, \infty)$. Their degrees are 4 and 12, and their lower/infinity variation counts are 2/2 and 6/6. In particular,

$$G_c(3/20) = \frac{73916290896185183967}{6400000000000000000000} > 0. \quad (4.5)$$

The non-importing implementation reconstructs all coefficients with `Fraction`, compactifies $x = 3/20 + y/(1 - y)$, and proves that every Bernstein coefficient is positive on a seven-interval dyadic cover. It uses neither SymPy nor the primary artifact.

For a non-root row, R-155 already supplies the independently certified bounds

$$L(y) < \frac{1}{30}, \quad E(y) < \frac{1}{1000} \quad (y \geq 27/20), \quad E(z) < \frac{1}{6} \quad (z \geq 3/20). \quad (4.6)$$

Thus

$$L(y) + 4E(y/9) + \frac{1}{4}E(y) < \frac{1}{30} + \frac{4}{6} + \frac{1}{4000} = \frac{2801}{4000} < \frac{25}{32}. \quad (4.7)$$

Equations (2.10)–(4.7), source payment (2.11), and the global PSD sextic form give

$$\frac{9}{10} - \frac{25}{32} = \frac{19}{160} > \frac{1}{10}. \quad (4.8)$$

5. Exact side-16 shell split and the 4/25 gap

For an actual nonzero side-16 dual mode,

$$x = \frac{\pi^2 N}{64}, \quad N = |n|^2 \in \mathbb{N}. \quad (5.1)$$

The primary verifier uses the exact Machin identity

$$\frac{\pi}{4} = 4 \arctan \frac{1}{5} - \arctan \frac{1}{239} \quad (5.2)$$

with alternating rational remainders. The independent verifier instead uses

$$\frac{\pi}{4} = \arctan \frac{1}{2} + \arctan \frac{1}{3}. \quad (5.3)$$

Their tangent identities equal one on the principal branch, and both exact enclosures imply

$$\frac{157}{50} < \pi < \frac{22}{7}. \quad (5.4)$$

It follows that

$$N = 1 \implies \frac{77}{500} < x < \frac{31}{200}, \quad N \geq 2 \implies x > \frac{3}{10}. \quad (5.5)$$

Set $t_L = 37/50$ and define G_L, P_L by (4.2)–(4.3) with t_c replaced by t_L . Exact Sturm chains prove both polynomials positive on the finite interval $[77/500, 31/200]$ and on $[3/10, \infty)$. The degree-4 sign guard has variation counts 3/3 and 1/1; the degree-12 squared polynomial has counts 7/7 and 5/5. Every interval contains zero roots and has positive endpoint data. The independent direct-interval and compactified Bernstein covers are also strictly positive. Therefore every actual nonzero mode obeys

$$L(x) + \frac{1}{4}E(x) < \frac{37}{50}. \quad (5.6)$$

The interior value in (4.7) also satisfies

$$\frac{2801}{4000} < \frac{37}{50}. \quad (5.7)$$

Hence the actual-lattice result is

$$\frac{9}{10} - \frac{37}{50} = \frac{4}{25}. \quad (5.8)$$

6. Failed shortcuts and honest globalization boundary

The exact certificates record three reusable route decisions.

First, the unweighted edge payment is not enough for the sharper theorem. At $x = 3/20$, the independently cleared inequality shows

$$L(x) + E(x) > \frac{37}{50}. \quad (6.1)$$

This is a failure of that comparison route, not a negative direction of the production Hessian.

Second, the nonzero-mode floor is load-bearing for the present majorant. At $x = 1/36$, exact rational arithmetic gives

$$4E(1/36) > \frac{25}{32}. \quad (6.2)$$

Thus the same row allocation cannot be extended unchanged to all $x > 0$. Equation (6.2) is a floor-removal obstruction for this envelope method, not an action counterexample.

Third, cardinality-independent origin coercivity does not automatically give a cardinality-independent nonzero neighbourhood. R-159 supplies compactness for one pure-dyadic shifted-state chart, but a growing union introduces a growing covariance-jet family and possible connection terms. A quantitative uniform modulus, or a new compact form argument, is still missing.

The path estimate can be stated on finitely supported vectors of an abstract one-sided infinite comparison path, but that observation does not construct a cutoff-free endpoint, close the production quadratic form, or prove regulator removal. Those analytic objects must be supplied before any infinite-cutoff theorem is claimed.

7. Devil’s-advocate review

1. **Objection: R-160 silently restores R-154’s disjointness. DISMISSED.** The theorem imports R-155’s exact fixed-law source-target-reuse algebra. A target may later be a raw source. What remains excluded is shifted-state recursion or nonlinear/revisited feedback.
2. **Objection: forward and legal reverse require two edge payments. DISMISSED.** They are one block and its adjoint in the selfadjoint Hessian. The quadratic factor two is exactly the left side of (3.2).
3. **Objection: L, E prove a full endpoint operator-norm bound. VALID WORDING CORRECTION.** They prove only the adverse lower comparison (2.10). Positive endpoint eigenvalues are not bounded here.
4. **Objection: factor-three sparsity is assigned to the sextic form. DISMISSED AFTER SCOPE PINNING.** Sparsity is asserted only for the endpoint/loss Hessian. The complete sextic Hessian is dropped as one PSD form and may have dense mixed entries.
5. **Objection: the proof used the coarse scratch polynomial as the exact A1 symbol. DISMISSED AFTER REPAIR.** Equations (2.1)–(2.4) record the error and its exact correction. Both executables rebuild every new polynomial from the authority coefficients.
6. **Objection: decimal values of π select the lattice shells. DISMISSED.** The two executables use different exact arctangent identities and rational alternating-series remainders.
7. **Objection: a length-independent finite-path constant proves cutoff removal. UPHELD AS A BOUNDARY.** The comparison constant is uniform, but the infinite production form and endpoint have not been constructed or closed.
8. **Objection: the origin gap supplies a uniform nonlinear radius. UPHELD AS A BOUNDARY.** Uniform control of the growing union’s third and higher jets is absent. R-160 makes no nonzero-radius assertion.
9. **Objection: the larger gap selects BCC or validates a PDE. UPHELD AS PROHIBITED.** This is phase-neutral conditional Hessian mathematics. No morphology, vacuum, lattice, or effective equation is selected or excluded.

8. Reproduction, route decision, and Result footer

From the repository root, run

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/  
a13_classii_weighted_schur_growing_affine_root_union_origin_gap_boundary.py  
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/  
a13_classii_weighted_schur_growing_affine_root_union_origin_gap_boundary_  
independent.py  
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/  
a13_classii_weighted_schur_growing_affine_root_union_origin_gap_boundary_  
verify.py
```

R-160 materially closes the origin-only growing-cardinality part of the R-155 affine chart: the gap now exceeds $1/10$ uniformly for every finite admitted root union, including factor-two source-target reuse. It does not close T-050. The next direct theorem must address a cardinality-uniform nonzero neighbourhood or an actual realised-past/nonlinear/revisit production law. That direct route remains conditional on the production law; phase-neutral T-054 candidate falsification continues in parallel. Truth, not BCC or any preselected PDE, remains the selection criterion.

Result ID: A13-CLASSII-WEIGHTED-SCHUR-GROWING-AFFINE-ROOT-UNION-ORIGIN-GAP-BOUNDARY (R-160).

Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.

Precise statement: Continuous radial gap $>19/160$ above $x=3/20$; actual side-16 dual-lattice gap $>4/25$, uniformly over every finite R-155 affine source-target-reuse root union at the origin.

Scope: Fixed cutoff/floor, centered zero-past covariance-matched common real-even scalar regulator $|m| \leq 1$, exact nonaliased torus, fixed-law affine origin Hessian; source-target reuse allowed.

Dependencies: A1, A7, R-128, R-130, R-151, R-155, and Sections 1--3.

Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.

Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/a13_classii_weighted_schur_growing_affine_root_union_
origin_gap_boundary_verify.py

Expected output: primary, independent, and integrated verification PASS.

Falsification gate: Any authority, symbol, incidence, resonance, coefficient, Sturm/Bernstein, pi enclosure, row, PSD ownership, record, or PDF gate fails.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: No uniform nonzero radius, realised-past, shifted-state/nonlinear/revisit, cutoff/floor removal, T-050/A13, Nelson/measure, phase/PDE, or Sector-A result follows.

Next required action: Uniform nonlinear modulus or an actual selected production-law conditional Hessian; keep truth-first T-054 in parallel.